Gradient-based parameter estimation in linear and non-linear state space models

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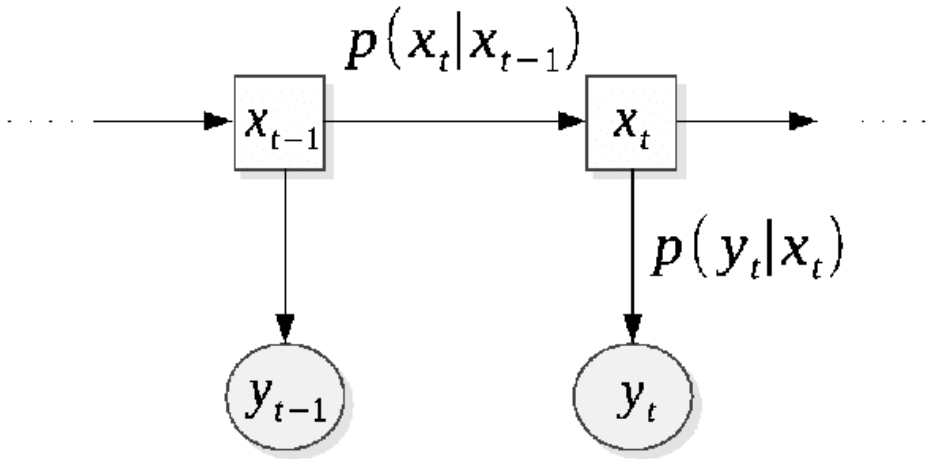
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State Space Models

Definiton

State Space Models (SSM) are a class of probabilistic graphical models that describe the probabilistic dependence between the latent state variable and the observed measurement.



Linear/Non-linear Gaussian Models

Linear gaussian state dynamic and observation model

$$\begin{aligned} x_{t+1} &= Ax_t + Bu_t + q_t & q_t &\sim \mathcal{N}(0, Q) \\ y_t &= Hx_t + r_t & r_t &\sim \mathcal{N}(0, R) \end{aligned}$$

Model parameters A, B, H, Q, R are supposed time invariant

Non-Linear gaussian state dynamic and observation model

$$\begin{aligned} x_{t+1} &= f(x_t, u_t) + q_t & q_t &\sim \mathcal{N}(0, Q) \\ y_t &= h(x_t) + r_t & r_t &\sim \mathcal{N}(0, R) \end{aligned}$$

Model functions f and h are supposed time invariant

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Kalman Filtering

The Kalman Filter is a recursive filter that estimates (via a joint probability distribution) the internal states $(x_{1:T})$ of a dynamic system from a series of noisy measurements $(y_{1:T})$.
Variables are assumed gaussian: $x_t \sim \mathcal{N}(\mu_t, \Sigma_t)$

Prediction Step (a-priori)

$$\hat{\mu}_{t+1} = A\mu_t + Bu_t$$
$$\hat{\Sigma}_{t+1} = A\Sigma_t A^T + Q$$

Correction Step (a-posteriori)

$$v_t = y_t - H\hat{\mu}_t$$
$$S_t = H\hat{\Sigma}_t H^T + R$$
$$K_t = \hat{\Sigma}_t H^T S_t^{-1}$$
$$\mu_t = \hat{\mu}_t + K_t v_t$$
$$\Sigma_t = (I - K_t H)\hat{\Sigma}_t$$

Kalman Filter as linear interpolation

A more intuitive visualisation of the updated state estimate:

$$\mu_{t+1} = (I - K_t H)\hat{\mu}_t + K_t y_t$$

The Kalman gain K_t chooses if the updated state depends more on the measurement or on the prediction

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Parametrized Filtering

Incentive:
Sometimes, the dynamics of the model are unknown or not fully known.

Linear parametrized gaussian state dynamic and observation model

$$x_{t+1} = A_\theta x_t + B_\theta u_t + q_t \qquad q_t \sim \mathcal{N}(0, Q_\theta)$$
$$y_t = H_\theta x_t + r_t \qquad r_t \sim \mathcal{N}(0, R_\theta)$$

Non-Linear parametrized gaussian state dynamic and observation model

$$x_{t+1} = f_\theta(x_t, u_t) + q_t \qquad q_t \sim \mathcal{N}(0, Q_\theta)$$
$$y_t = h_\theta(x_t) + r_t \qquad r_t \sim \mathcal{N}(0, R_\theta)$$

Model parameters θ are supposed time invariant.

Maximum A-Posteriori (MAP) Estimate

$\theta \in \mathbb{R}^d$: unknown model parameters to be to be estimated.
Bayesian approach $\rightarrow$ prior on the unknown parameters: $\theta \sim p(\theta)$.

For a set of measurements $y_{1:T} = \{y_1, ..., y_T\}$, the aim is to learn the posterior:

$$p(\theta|y_{1:T}) = \frac{p(y_{1:T}|\theta)p(\theta)}{p(y_{1:T})}$$

However it cannot be obtained in closed form, except for linear cases.

Maximum A-Posteriori (MAP) Estimate

MAP Estimate:

By definition:

$$\theta_{MAP} = \underset{\theta}{\text{argmax}}\, \mathcal{L}_{y_{1:T}}(\theta)$$

Using the Laplace approximation on the posterior:

$$p(\theta|y_{1:T}) \approx \mathcal{N}(\theta|\theta^{MAP}, \boldsymbol{E}^{-1})$$

where $\boldsymbol{E}$ is the Hessian of the likelihood $\mathcal{L}$ evaluated at θ^{MAP}. We can thus obtain an approximate on the energy likelihood[1]:

$$\mathcal{L}_{y_{1:T}}(\theta) \approx \frac{1}{2}\sum_{t=1}^{T}[v_t^T(\theta)S_t^{-1}(\theta)v_t(\theta) + log(|S_t(\theta)|)] - log(p(\theta))$$

[1] Simo Särkkä. *Bayesian Filtering and Smoothing*. English. United Kingdom: Cambridge University Press, 2013. ISBN: 9781107619289.

Expectation Maximization (EM) Algorithm

For a linear gaussian model: $\theta = \{A, B, H, Q, R\}$.
The log likelihood of the complete data is given by:[2]

$$\mathcal{L}_{x_{1:T}, y_{1:T}} = \mathcal{L}_{x_{1:T}|y_{1:T}} + \mathcal{L}_{y_{1:T}}$$

The EM algorithm consists of iteratively:

1. *E-step*: computing $\mathcal{Q}(\theta, \theta_j) = \mathbb{E}[\mathcal{L}_{x_{1:T}, y_{1:T}}, \theta_j]$
2. *M-step*: maximizing $\mathcal{Q}(\theta, \theta_j)$, equivalent to maximizing $\mathcal{L}_{y_{1:T}}$

The M-step is obtained by actually solving the system of equations for a given parameter. The equations for a linear model are easy, but can be totally intractable for non-linear cases.

[2] Javier Cara, Jesus Juan, and E Alarcón. "Using the EM algorithm to estimate the state space model for OMAX". In: Sept. 2014.

Gradient-based parameter estimation

$\nabla_\theta \mathcal{L}(\theta)$ can be easily computed using automatic differentiation (AD).
$\rightarrow$ "Black box" solution that is model independent.

Algorithm 1: MAP estimate using gradient descent for n epochs

Result: $\theta_n \approx \theta_{MAP}$

$\theta_0 = \theta_{init}$;

for *i in 1:n* **do**

  run filter with θ_i for T states;

  compute $\nabla_{\theta_i} \mathcal{L}(\theta_i)$ with updated states;

  $\theta_{i+1} \leftarrow \theta_i$;

end

ADAM optimizer was used, however any optimiser algorithm can be used.

Results: Simple Linear Model (1)

$$x_t = \begin{pmatrix} p \\ v \\ a \end{pmatrix}_t = \begin{pmatrix} x \\ \dot{x} \\ \ddot{x} \end{pmatrix}, u_t = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Ground truth model (discretized $\Delta t = 0.001$):

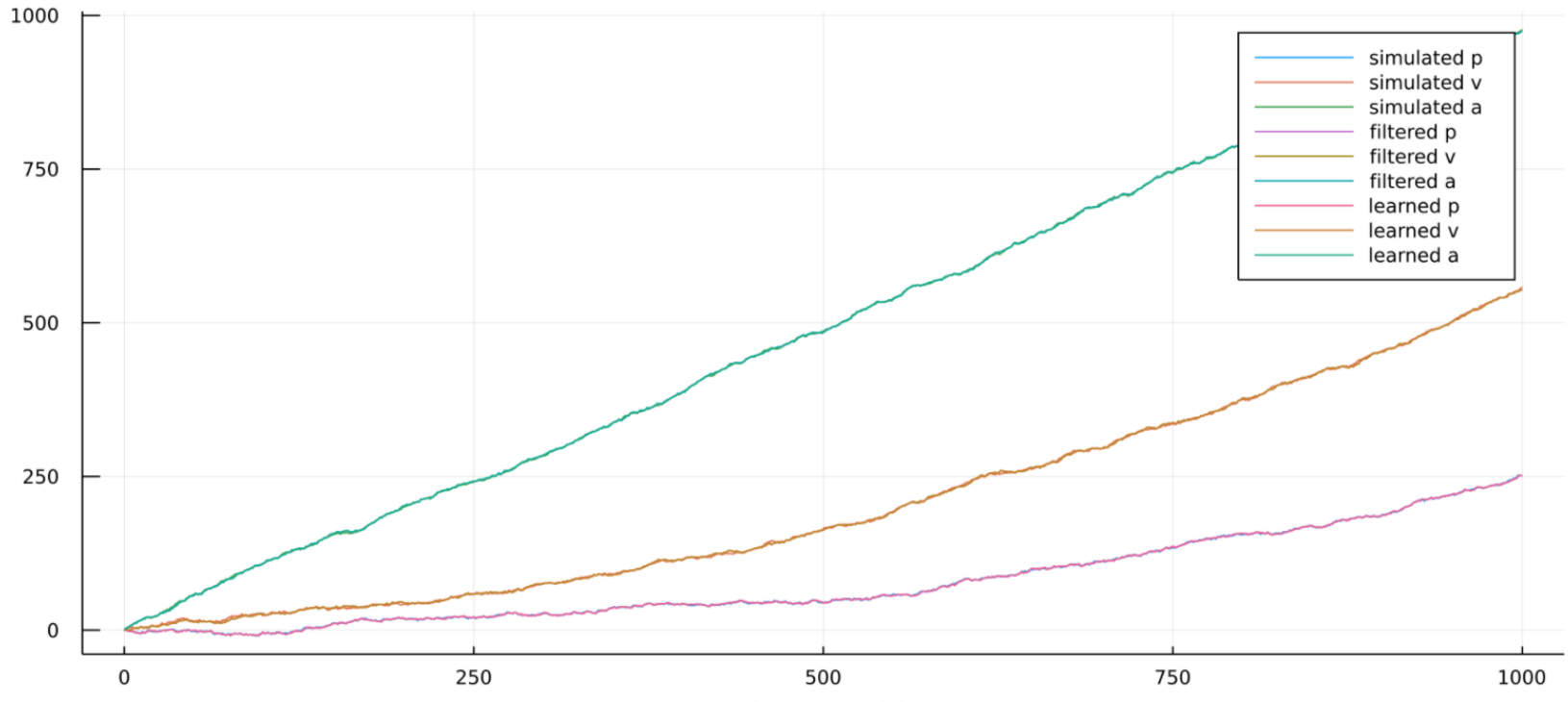
$$A = \begin{pmatrix} 1 & \Delta t & \frac{1}{2}\Delta t^2 \\ 0 & 1 & \Delta t \\ 0 & 0 & 1 \end{pmatrix}, B = \begin{pmatrix} 0 & 0 & 1 \end{pmatrix}, Q = I_3$$

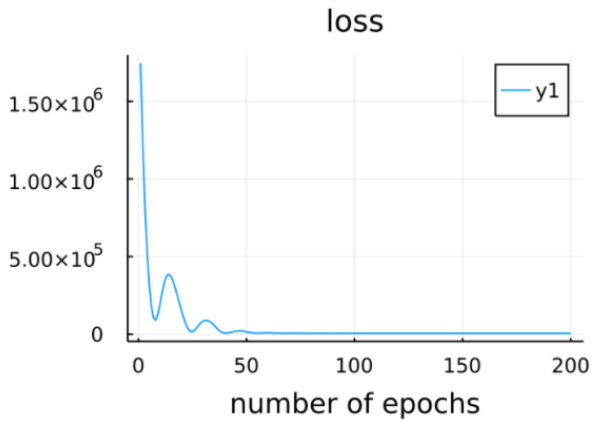
$$H = randn(3,3), R = I_3$$

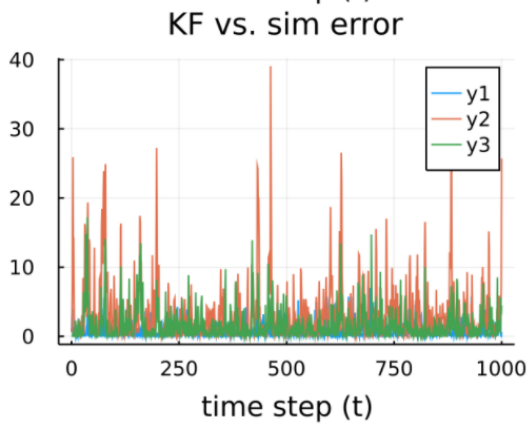
Parametrized model:

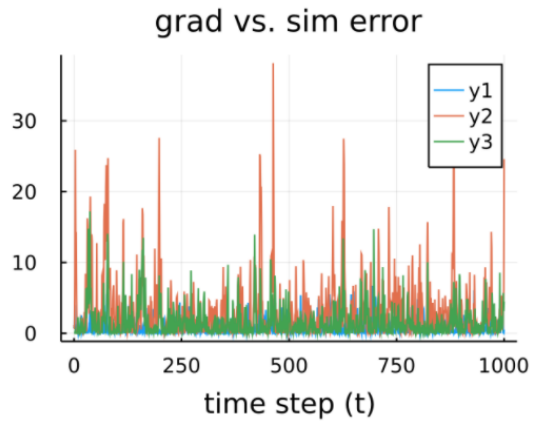
$$A_\theta = \begin{pmatrix} \theta_1 & \theta_2 & \theta_3 \\ 0 & \theta_1 & \theta_2 \\ 0 & 0 & \theta_1 \end{pmatrix}$$

Results: Simple Linear Model (2)

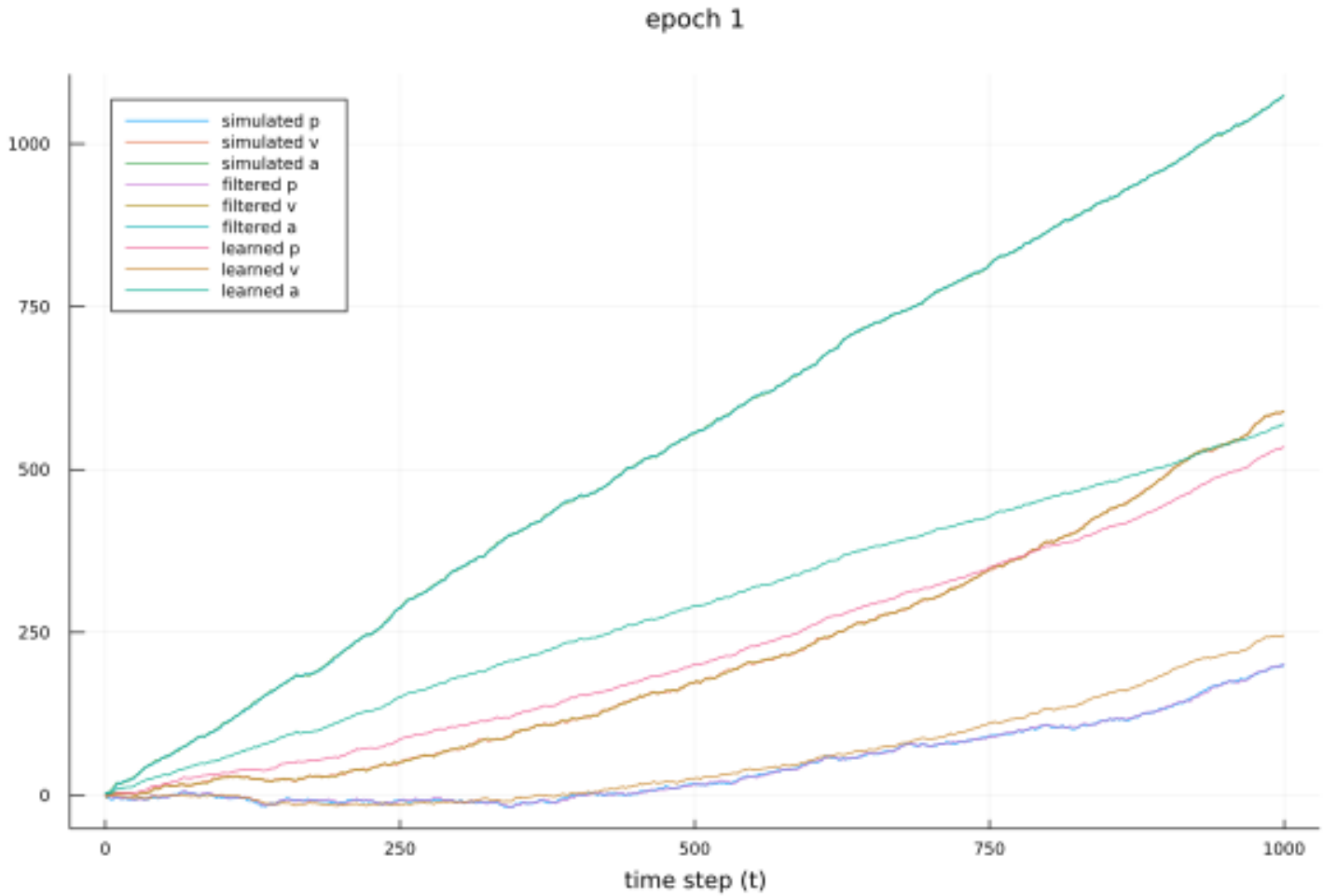






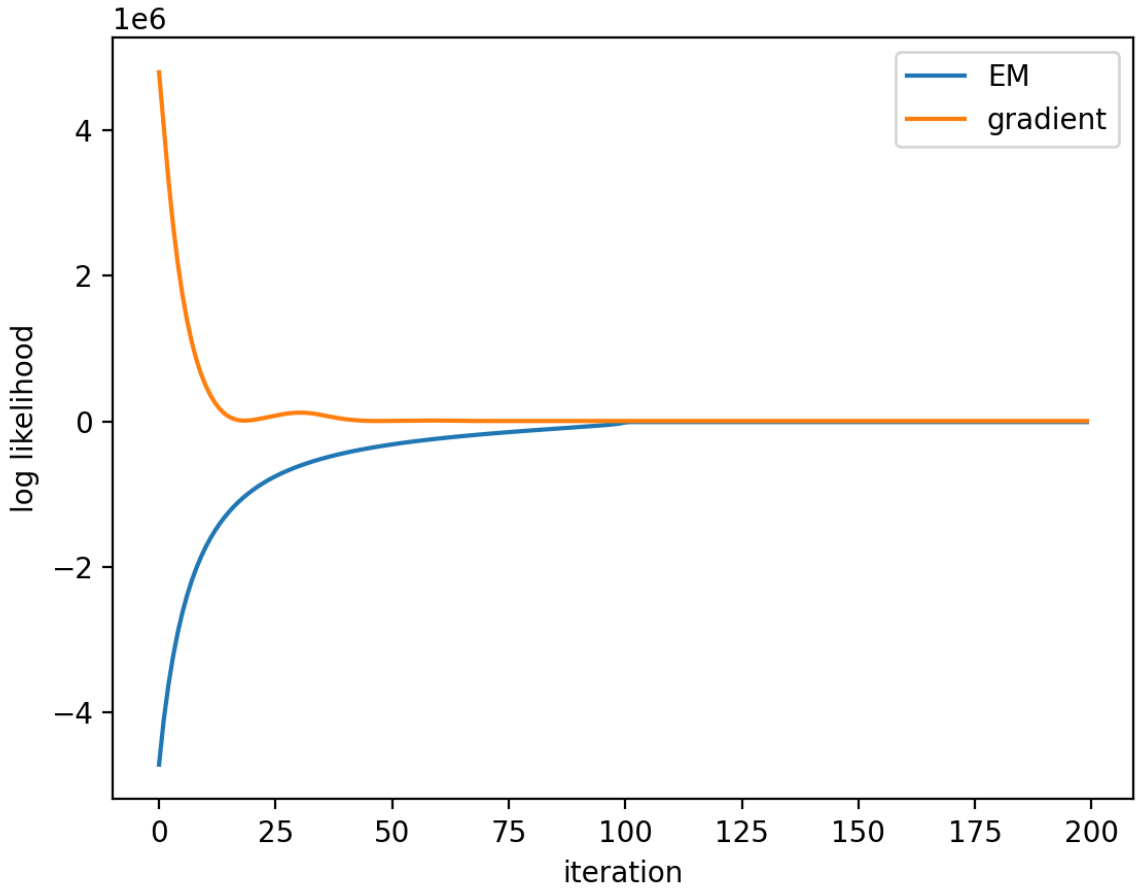


Results: Simple Linear Model (3)



Results: Simple Linear Model (4)

Gradient-based estimation vs. EM algorithm Initialisation:
$A_0 = A + 0.5 * randn(3, 3)$



Results: Simple Pendulum (1)

$$x_t = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}_t = \begin{pmatrix} \alpha \\ \dot{\alpha} \end{pmatrix}_t, u_t = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

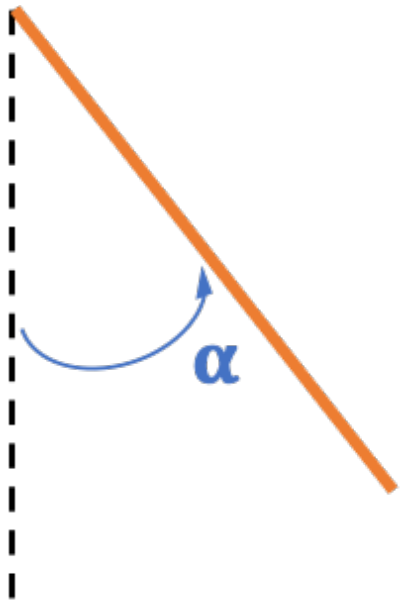
Ground truth model:

$$f(x,u) = \begin{pmatrix} x_1 + x_2 \Delta t \\ x_2 - \frac{g}{L} sin(x_1) \Delta t \end{pmatrix}, Q = I_2$$

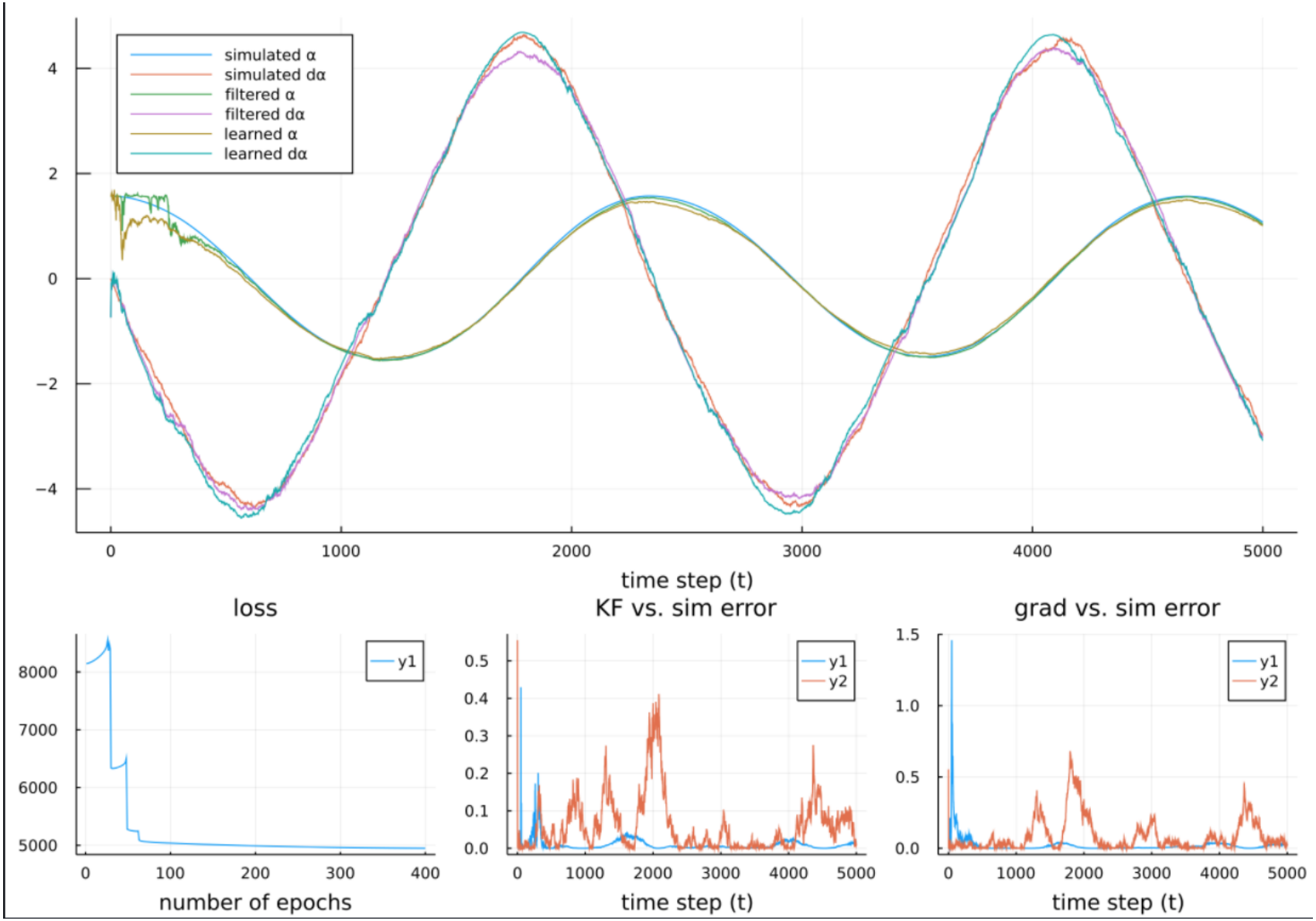
$$h(x) = \begin{pmatrix} sin(x_1) \\ sin(x_2) \end{pmatrix}, R = I_2$$

Parametrized model:

$$f_\theta(x,u) = \begin{pmatrix} x_1 + \theta_1 x_2 \Delta t \\ x_2 - \theta_2 sin(\theta_3 x_1) \Delta t \end{pmatrix}$$



Results: Simple Pendulum (2)



Results: Simple Pendulum (3)

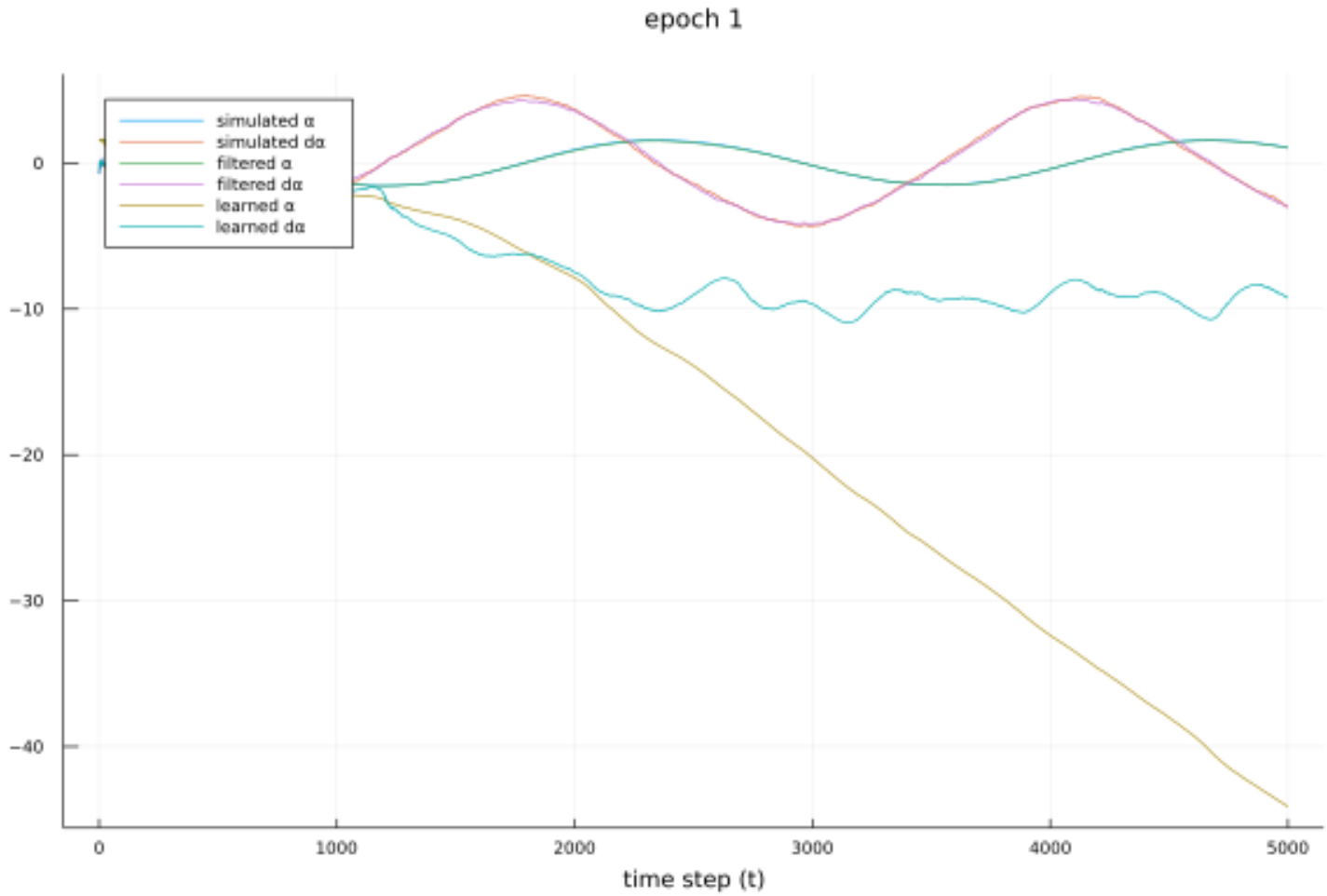


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Discussion

- Gradient-based estimation in line with "black box" philosophy of AD
- Precise parameter selection
- Switch from linear to non-linear gradient estimation is seamless
- Faster convergence than EM (only tested on linear models)
- Next step: implement f_θ as MLP

References

Cara, Javier, Jesus Juan, and E Alarcón. "Using the EM algorithm to estimate the state space model for OMAX". In: Sept. 2014.

Särkkä, Simo. *Bayesian Filtering and Smoothing*. English. United Kingdom: Cambridge University Press, 2013. ISBN: 9781107619289.